

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO4: Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization.(BL3,BL4,BL5)

Assessment Tool Weightage: 40%

Dr G. A. SENTHIL

QUESTIONS: 25

Total Marks: 50

Annexure A

Design and Develop Model

DURATION: 4 HRS (2 Sessions)

Code of Conduct:

I Vignesh S / 192325002 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Vignesh S

DESIGN AND DEVELOP MODEL — ANSWERS

One question answered from each of the five sections, each presented as a diagram with a supporting explanation.

Section 1 • Model-Based RL Approach

Question: Draw a flowchart illustrating the complete workflow of a Model-Based Reinforcement Learning agent from environment interaction to policy optimization.

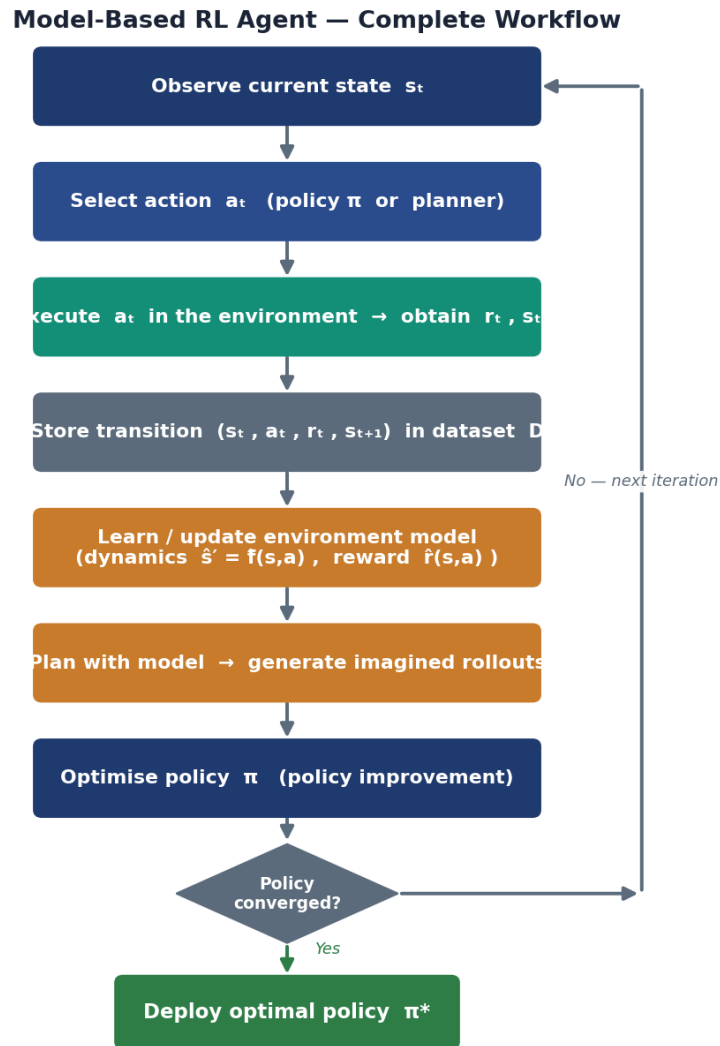


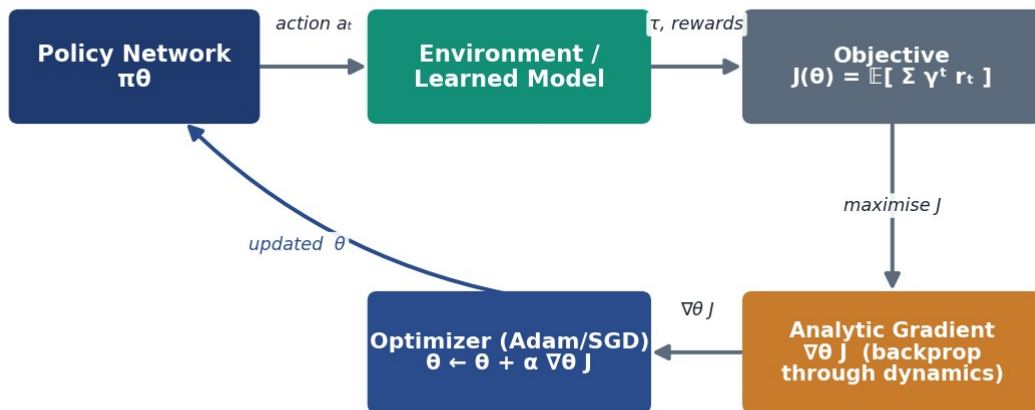
Figure 1: Model-Based RL Approach — flowchart.

Explanation: The flowchart traces one complete iteration of a model-based RL agent. The agent observes the current state s_t and selects an action a_t using its current policy or a model-based planner. Executing a_t in the environment yields a reward r_t and next state s_{t+1} , and the transition (s_t, a_t, r_t, s_{t+1}) is stored in a dataset. From this data the agent learns an environment model — a dynamics function $\hat{f}$ that predicts the next state and a reward model $\hat{r}$. The learned model is then used to plan: it generates imagined rollouts that let the agent evaluate actions without touching the real environment, and these are used to optimise the policy π (policy improvement). The loop repeats, and once the policy converges the optimal policy π^* is deployed. The essential idea is the closed loop of act $\rightarrow$ learn model $\rightarrow$ plan $\rightarrow$ improve, which reuses real experience and greatly improves sample efficiency.

Section 2 • Analytic Gradient Computation

Question: Draw a block diagram illustrating gradient computation, parameter updates, and reward optimization in Reinforcement Learning.

Analytic Gradient Computation — Parameter Updates & Reward Optimisation



Gradients of expected return flow back through the differentiable model to update the policy parameters each step.

Figure 2: Analytic Gradient Computation — block diagram.

Explanation: This block diagram shows how the policy parameters θ are optimised using analytic gradients. The policy network π_θ outputs actions that drive the environment or a differentiable learned model, producing a trajectory τ and its rewards. These define the objective $J(\theta)$ — the expected discounted return the agent seeks to maximise. Because the model and policy are differentiable, the gradient $\nabla_\theta J$ can be computed analytically by backpropagating the return through the dynamics and the policy; this pathwise (analytic) gradient is typically lower-variance than score-function estimates. The optimizer (SGD/Adam) then updates the parameters, $\theta \leftarrow \theta + \alpha \nabla_\theta J$, ascending the objective, and the updated θ flows back to the policy network. Repeating this loop performs gradient-based reward optimisation, steadily improving the policy toward higher return.

Section 3 · Sampling-Based Planning

Question: Draw a flowchart illustrating the sampling-based planning process in Model-Based Reinforcement Learning.

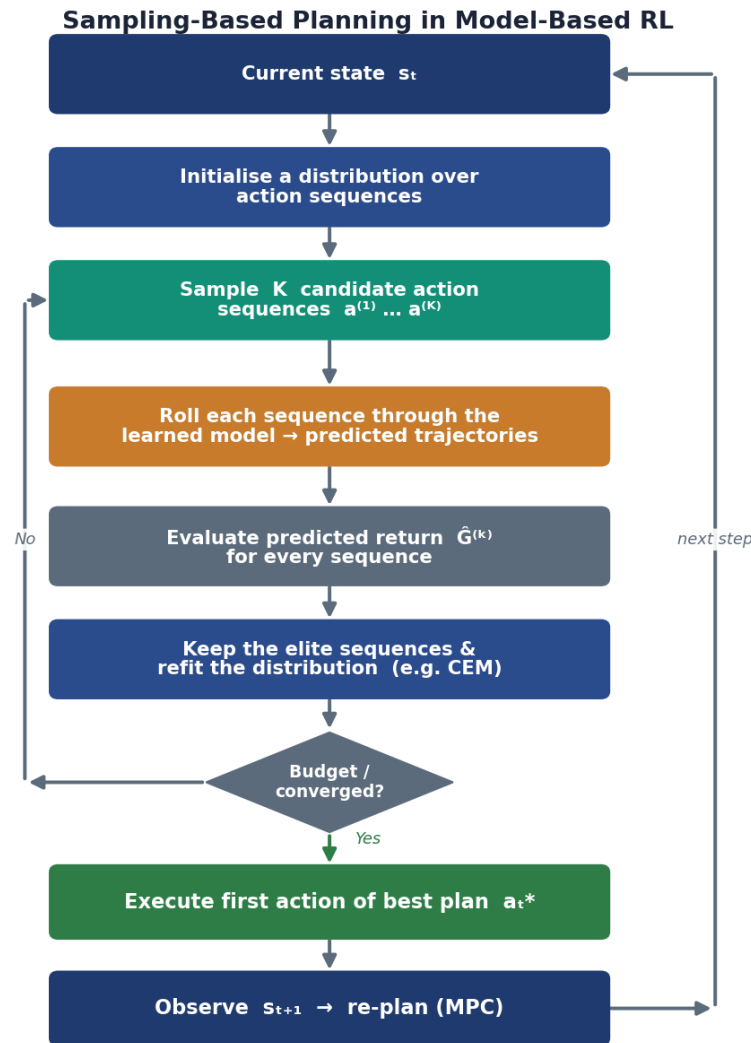


Figure 3: Sampling-Based Planning — flowchart.

Explanation: Sampling-based planning selects actions by simulating many possibilities in the learned model instead of searching exhaustively. Starting from the current state s_t , the planner initialises a distribution over action sequences and samples K candidate sequences. Each sequence is rolled out through the learned model to produce a predicted trajectory and its return $\hat{G}$. The best (elite) sequences are retained and the sampling distribution is refit toward them — the core of the Cross-Entropy Method (CEM). This sample–evaluate–refit cycle repeats until a computational budget or convergence criterion is reached. The first action of the best plan is executed in the real environment; the agent then observes the new state and re-plans, following the receding-horizon principle of Model Predictive Control (MPC). Sampling-based planning scales naturally to large or continuous action spaces, where exhaustive search would be infeasible.

Section 4 • Model-Based Data Generation

Question: Prepare a concept map showing the relationship among environment model, generated experiences, replay buffer, and policy learning.

Model-Based Data Generation — Concept Map

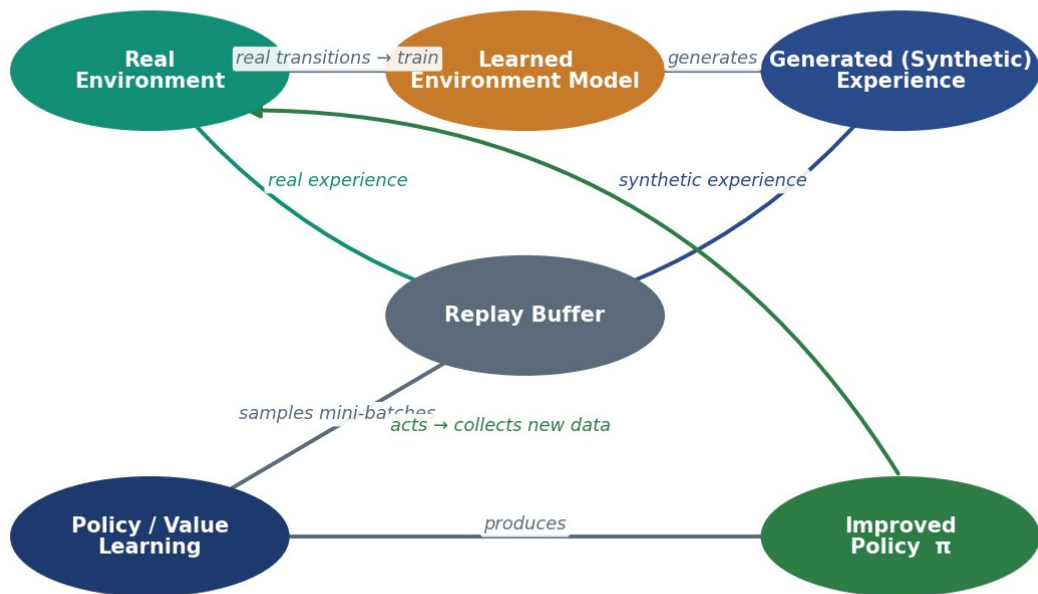


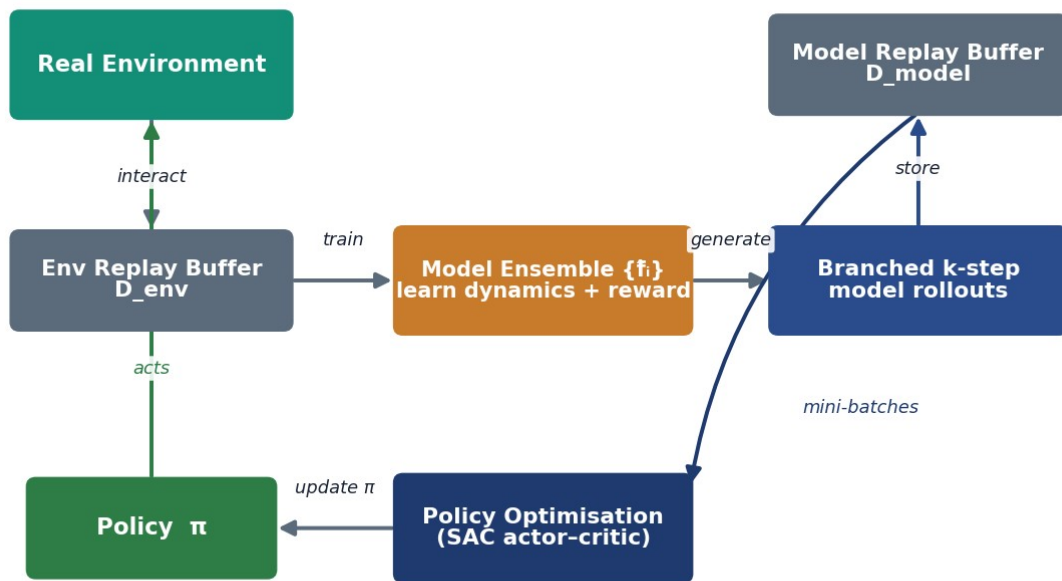
Figure 4: Model-Based Data Generation — concept map.

Explanation: The concept map shows how a learned model is used to manufacture additional training data. The agent collects real transitions from the environment and uses them to train the learned environment model. Once trained, the model generates synthetic (imagined) experience. Both real and synthetic experiences are stored together in a replay buffer, from which mini-batches are sampled to train the policy and value functions. The resulting improved policy then acts in the environment to collect fresh real data, closing the loop. By augmenting scarce real data with cheap model-generated data, the agent learns from far more experience than the environment alone could provide — the essence of Dyna-style model-based RL. This markedly improves sample efficiency, provided the model is accurate enough that its synthetic data does not mislead learning.

Section 5 • Value-Equivalence Prediction and Policy Optimization

Question: Draw a block diagram illustrating the complete Model-Based Policy Optimization (MBPO) framework.

Model-Based Policy Optimization (MBPO) Framework



Short model-generated rollouts branched from real states expand data cheaply, boosting sample efficiency.

Figure 5: Value-Equivalence Prediction and Policy Optimization — block diagram.

Explanation: Model-Based Policy Optimization (MBPO) combines a learned model with an off-policy actor–critic. The agent interacts with the real environment and stores real transitions in the environment replay buffer D_{env} . An ensemble of models is trained on D_{env} to capture the dynamics and reward and to represent model uncertainty. Crucially, MBPO generates only short, k -step rollouts branched from real states sampled from D_{env} , and stores them in a separate model replay buffer D_{model} . The policy is then optimised — for example with Soft Actor–Critic (SAC) — mainly on mini-batches drawn from D_{model} , and the improved policy acts again in the real environment. Keeping the rollouts short limits compounding model error, while the large volume of cheap model data makes the method highly sample-efficient. MBPO is a strong, widely-cited template for model-based policy optimisation.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools,	Appropriate	Limited use of	Minimal or

		well-analyzed data, and highly effective visuals	tools and data used effectively with clear visuals	tools and basic data interpretation	incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism ; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness